

Refresher Course in Calculus, Probability, and Statistics

Day 3: Probability

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Introduction

"Life's most important questions are, for the most part, nothing but probability problems."

Pierre Simon Laplace

"Doubt is not a pleasant condition, but certainty is absurd."

Voltaire

- ▶ Most aspects of the world around us have an element of uncertainty:
 - Will it rain tomorrow?
 - Will I win the lottery next week?
 - Will I earn more than 6,000 CHF/month some day?
- ▶ Theory of probability provides mathematical tools for quantifying and describing this randomness and dealing with uncertainty.
- ▶ Probability is needed to understand regression analysis and statistics.

Probability

Basic definitions

- ▶ **Trial:** event whose outcome is unknown (also called: experiment, observation)
 - flipping a coin
 - rolling a die
- ▶ **Sample space:** specification of all possible outcomes of a trial
 - for flipping a coin: $S = \{\text{heads}, \text{tails}\}$
 - for rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$
- ▶ **Outcome:** mutually exclusive potential result of a random trial
- ▶ **Event:** specification of the outcome of one trial (single outcome or set of outcomes)
 - event “heads in flipping a coin”: $A = \{\text{heads}\}$
 - event “odd number in rolling a die”: $B = \{1, 3, 5\}$
- ▶ **Mutual exclusivity:** events that cannot occur together are mutually exclusive (e.g. passing a test and not passing a test)
- ▶ **Independence:** the outcome of one trial has no relationship with the outcome of another trial

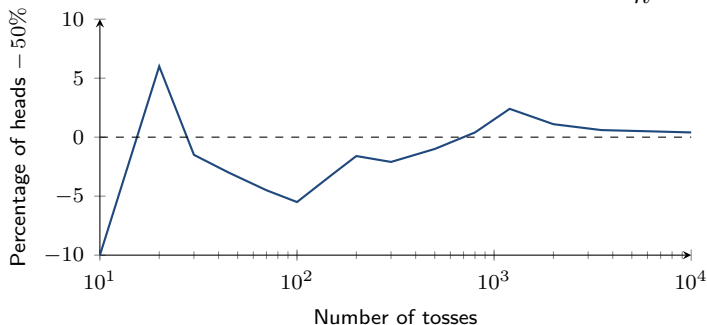
Probability

Basic definitions (continued)

► Relative frequency definition of probability:

If an experiment is repeated n times under essentially identical conditions and the event A occurs m times, then as n gets large the ratio $\frac{m}{n}$ approaches the probability of A :

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}.$$



Kerrich's experiment: tossing a coin 10,000 times.

Probability

Properties

Event	Probability
A	$P(A) \in [0, 1]$
not A	$P(A^c) = P(\bar{A}) = 1 - P(A)$
A or B	$P(A \cup B) = P(A) + P(B)$ if A, B mutually exclusive
A or B	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
A and B	$P(A \cap B) = P(A) P(B)$ if A, B independent
A and B	$P(A \cap B) = P(A B) P(B) = P(B A) P(A)$
A given B	$P(A B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B A) P(A)}{P(B)}$
Bayes' theorem	

Random Variables

- ▶ **Random variable** (or stochastic variable): variable whose value results from measurement of a random process
 - numerical summary of a random process
 - probabilities are associated with the possible values of the random variable
- ▶ Examples of a random variable:
 - number of times Word crashes while writing a term paper
 - number of times heads comes up when tossing a coin 10 times
 - length of time it takes a random student to understand Bayes' theorem
- ▶ Random variables can be:
 - **Discrete:** can take only a limited number of values (gender, outcome of a die, ...)
 - **Continuous:** can take any value in an interval (time, earnings, ...)

Discrete Random Variables

Probability Density Function

- ▶ **Probability density function (pdf):** summarizes the probabilities of outcomes
- ▶ For **discrete random variables** it is like a table contrasting the possible values of the variable with the probability that each value occurs
 - indicates the probability of each possible value occurring
 - the value of the pdf of a discrete random variable X , denoted $f(x)$, is the probability that X takes the value x :

$$f(x) = P(X = x)$$

- $0 \leq f(x) \leq 1$
- if X takes n possible values $x_1, \dots, x_n$:

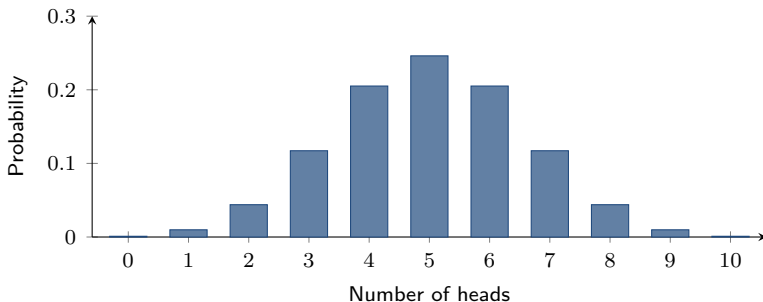
$$P(X = x_1) + \dots + P(X = x_n) = f(x_1) + \dots + f(x_n) = 1$$

Discrete Random Variables

Probability Density Function: Example

► pdf for the number of heads out of 10 coin tosses:

Outcome (# heads)	0	1	2	3	4	5	6	7	8	9	10
$f(x)$	.0010	.0098	.0439	.1172	.2051	.2461	.2051	.1172	.0439	.0098	.0010



Discrete Random Variables

Probability Density Function: Example (continued)

- The probability of an event (or compound event) can be computed:
 - probability of event “exactly 7 heads”:

$$P(X = 7) = 0.1172 = 11.72\%$$

- probability of compound event “even number of heads”:

$$\begin{aligned} &P(X=0) + P(X=2) + P(X=4) + P(X=6) + P(X=8) + P(X=10) \\ &= 0.0010 + 0.0439 + 0.2051 + 0.2051 + 0.0439 + 0.0010 = 0.5000 = 50\% \end{aligned}$$

- probability of compound event “no more than 3 heads”:

$$\begin{aligned} &P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= 0.0010 + 0.0098 + 0.0439 + 0.1172 = 0.1719 = 17.19\% \end{aligned}$$

Discrete Random Variables

Cumulative Density Function

- ▶ **Cumulative distribution function (cdf):** an alternative way to summarize probabilities
- ▶ the cdf of a random variable X , denoted $F(x)$, is the probability that X is less than or equal to x :

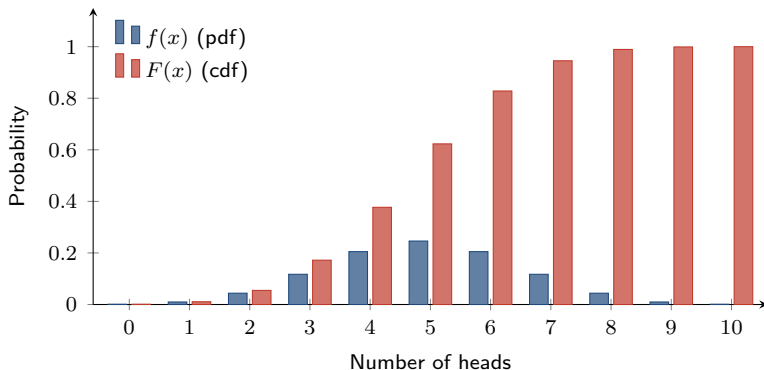
$$F(x) = P(X \leq x)$$

- ▶ cdf for the number of heads out of 10 tosses:

# heads	0	1	2	3	4	5	6	7	8	9	10
$f(x)$	.0010	.0098	.0439	.1172	.2051	.2461	.2051	.1172	.0439	.0098	.0010
$F(x)$	.0010	.0107	.0547	.1719	.3770	.6230	.8281	.9453	.9893	.9990	1.000

Discrete Random Variables

Cumulative Density Function (continued)



Discrete Random Variables

Special Case: Binary Variables

- ▶ Important special case of a discrete random variable: one that can take only two possible values (0 or 1)
 - such variables are called *binary*, *indicator*, *dummy*, or *dichotomous* variables
- ▶ Probability distribution of a binary random variable is governed by a **Binomial** or **Bernoulli distribution**
 - a sequence of n independent Bernoulli trials with constant probability of success p at each trial; we are interested in the total number of successes x
- ▶ **Example:** in the 4th quarter of 1988 in Massachusetts, 60 newborns tested positive for HIV antibodies.
 - what are the chances that 30 children will be infected?
 - what are the chances that 30+ are infected?
 - possible model: binomial with $p = 0.25$ and $n = 60$



Jakob Bernoulli
1655–1705

Discrete Random Variables

Special Case: Binary Variables (continued)

► The **Binomial distribution**:

$$P(x) = \underbrace{\binom{n}{x}}_{\text{binomial coefficient}} p^x (1-p)^{n-x} = \underbrace{\frac{n!}{x!(n-x)!}}_{\text{binomial coefficient}} p^x (1-p)^{n-x}$$

- x the number of successes
- p the probability of success
- n the number of trials

► Example (continued):

- Probability that 30 newborns will be HIV-positive:

$$P(30) = \binom{60}{30} 0.25^{30} (1-0.25)^{60-30} = \frac{60!}{30!30!} \cdot 0.25^{30} \cdot 0.75^{30} \approx 1.83 \times 10^{-5} = 0.0018\%$$

- So the probability that 30 or more newborns will be HIV-positive:

$$P(x \geq 30) = 1 - P(x < 30) = 1 - \sum_{i=0}^{29} P(x = i) \approx 2.67 \times 10^{-5} = 0.0027\%$$

- ### ► A good visualization of the logic of the binomial distribution is the *quincunx*: <http://www.mathsisfun.com/data/quincunx.html>

Discrete Random Variables

Expected Value

- ▶ **Mathematical expectation:** the mean of a random variable (its long-run average value over many repeated trials or occurrences)
- ▶ Expected value of a discrete random variable X taking values $x_1, \dots, x_n$:

$$E(X) = \mu_X = x_1 P(X = x_1) + \dots + x_n P(X = x_n) = \sum_{i=1}^n x_i f(x_i) = \sum_x x f(x)$$

→ a weighted average of the possible outcomes (weights = probabilities)

- ▶ For a **binary variable**:

$$E(B) = 1 \times p + 0 \times (1 - p) = p$$

→ for the binomial distribution with n trials:

$$E(B) = \mu_B = np$$

- ▶ For **linear combinations**:

$$E(aX + bY + c) = a E(X) + b E(Y) + c$$

Discrete Random Variables

Variance and Standard Deviation

- **Variance** and **standard deviation** measure the dispersion or “spread” of a probability distribution

- **larger** variance and standard deviation
⇒ more “spread out” values

- **Variance** of a discrete random variable X :

$$\text{var}(X) = \sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - \mu_X^2$$

- For a **binary variable**:

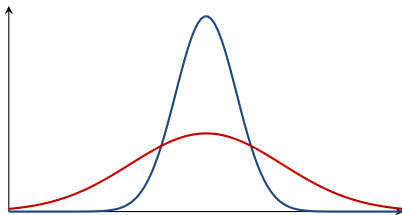
$$\text{var}(B) = p(1 - p)$$

- For the **binomial distribution** with n trials:

$$\text{var}(B) = np(1 - p)$$

- For **linear combinations**:

$$\text{var}(aX + bY + c) = a^2 \text{var}(X) + b^2 \text{var}(Y) + 2ab \text{cov}(X, Y) = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \sigma_{XY}$$



Discrete Random Variables

Variance and Standard Deviation (continued)

- ▶ **Standard deviation:** $\sigma_X = \sqrt{\text{var}(X)} = \sqrt{\sigma_X^2}$
 - same units of measure as the random variable
- ▶ For our example with the newborns:
 - how many should we expect to be HIV-positive?

$$E(B) = \mu_B = np = 60 \times 0.25 = 15$$

- apply an empirical rule that $\approx 2/3$ of a (symmetric) distribution lies in the range $\mu_B \pm \sigma_X$ and 95% is covered by $\mu_B \pm 2\sigma_X$

$$\sigma_B = \sqrt{\text{var}(B)} = \sqrt{np(1-p)} = \sqrt{60 \times 0.25 \times (1-0.25)} = 3.354$$

- $15 \pm 3.3541 = [11.65, 18.35]$
 - $15 \pm 6.7082 = [8.29, 21.71]$
 - NB: in our case the distribution is not perfectly symmetric
- more exact intervals are $[11.56, 18.99]$ and $[8.83, 22.72]$

Discrete Random Variables

Other Measures of the Shape of a Distribution

- **Skewness** measures lack of symmetry:

$$\text{Skewness} = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$$

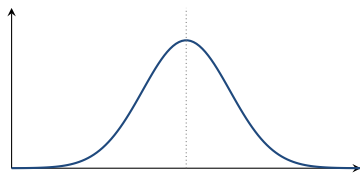
- $S = 0$: symmetric; $S < 0$: left-skewed (long left tail); $S > 0$: right-skewed (long right tail)
- skewness is unitless

- **Kurtosis** measures how much mass is in the tails of a distribution:

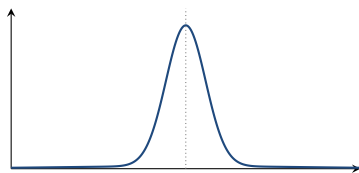
$$\text{Kurtosis} = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}$$

- how much of the variance of X comes from extreme values
- benchmark value is 3 (normal distribution); $K - 3$ is called “excess kurtosis”
- $K = 3$: mesokurtic; $K < 3$: **platykurtic**; $K > 3$: **leptokurtic**
- kurtosis is unitless and cannot be negative

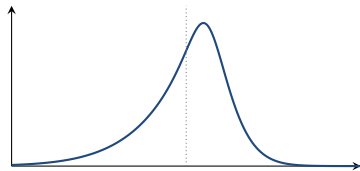
Four Distributions with Different Skewness and Kurtosis



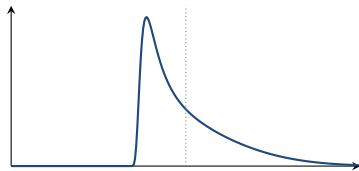
(a) Skewness = 0, kurtosis = 3



(b) Skewness = 0, kurtosis = 20



(c) Skewness = -1.0, kurtosis = 5



(d) Skewness = 1.4, kurtosis = 5

Note: (a),(b) are Gaussian, while (c),(d) are sinh-arcsinh (SHASH). A Gaussian forced to high levels of skewness would grow a non-convex bump in the tail, so SHASH is used for panels (c) and (d) to keep smooth, convex tails at the same kurtosis. The dotted line marks the mean ($x = 0$).

Several Discrete Random Variables

Joint and Marginal Distributions

- ▶ **Joint probability** that $X = x$ and $Y = y$ (joint pdf of X and Y):

$$f(x, y) = P(X = x, Y = y)$$

- Sum of all joint probabilities: $\sum_x \sum_y f(x, y) = 1$

- ▶ **Marginal probability distribution** of random variable X :

$$f_X(x) = \sum_y f(x, y) \quad \text{for each value of } X$$

- just another name for its probability distribution
- computed from the joint distribution of X and Y by adding up the probabilities of all outcomes for which X takes on a specific value

Several Discrete Random Variables

Conditional Distribution

- **Conditional pdf:** the probability that random variable Y takes the value y given that $X = x$:

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)} \quad \Longrightarrow \quad f(y \mid x) = \frac{f(x, y)}{f_X(x)}$$

- **Statistical independence:**

$$P(Y = y \mid X = x) = P(Y = y) \quad \Longrightarrow \quad f(y \mid x) = f(y) = f_Y(y)$$

Several Discrete Random Variables

Example

- ▶ Joint distribution of computer crashes (C) and operating system (OS):

Operating system	$C=0$	$C=1$	$C=2$	$C=3$	$C=4$	Total
Windows ($OS=0$)	0.531	0.08	0.05	0.029	0.01	0.70
Linux ($OS=1$)	0.269	0.02	0.01	0.001	0.00	0.30
Total	0.80	0.10	0.06	0.03	0.01	1.00

- ▶ Probability the computer does not crash at all *and* runs Linux:
Joint probability: $P(C=0, OS=1) = 0.269 = 26.9\%$
 - ▶ Probability a random computer runs Windows:
Marginal probability: $P(OS=0) = 0.531 + 0.08 + 0.05 + 0.029 + 0.01 = 0.70$
 - ▶ Probability a Linux computer does not crash:
Conditional probability: $P(C=0 | OS=1) = \frac{0.269}{0.30} = 0.8967 = 89.67\%$
- Under statistical independence this would be $P(C=0 | OS=1) = P(C=0) = 0.80$;
since $0.8967 \neq 0.80$, crashes and OS are *not* independent.

Several Discrete Random Variables

Conditional Expectation

- **Conditional expectation** (conditional mean) of Y given $X = x$, if Y can take k values $y_1, \dots, y_k$:

$$E(Y \mid X = x) = \sum_{i=1}^k y_i P(Y = y_i \mid X = x) = \sum_y y f(y \mid x)$$

- Example: how many crashes do I expect when working with each OS?
 - Linux computer:

$$E(C \mid OS=1) = 0 \cdot \frac{0.269}{0.30} + 1 \cdot \frac{0.02}{0.30} + 2 \cdot \frac{0.01}{0.30} + 3 \cdot \frac{0.001}{0.30} + 4 \cdot \frac{0.00}{0.30} \approx 0.143$$

- Windows computer:

$$E(C \mid OS=0) = 0 \cdot \frac{0.531}{0.70} + 1 \cdot \frac{0.08}{0.70} + 2 \cdot \frac{0.05}{0.70} + 3 \cdot \frac{0.029}{0.70} + 4 \cdot \frac{0.01}{0.70} \approx 0.439$$

Several Discrete Random Variables

Conditional Variance

- **Conditional variance** (variance of Y given a value for X):

$$\text{var}(Y \mid X = x) = \sum_{i=1}^k \left[(y_i - E(Y \mid X = x))^2 P(Y = y_i \mid X = x) \right]$$

- Example: conditional variance of crashes for each OS

- Linux computer:

$$\begin{aligned} \text{var}(C \mid OS=1) &= \sum_{i=1}^4 \left[(c_i - E(C \mid OS=1))^2 P(C = c_i \mid OS=1) \right] \\ &= (0 - 0.143)^2 \times \frac{0.269}{0.30} + (1 - 0.143)^2 \times \frac{0.02}{0.30} + (2 - 0.143)^2 \times \frac{0.01}{0.30} \\ &\quad + (3 - 0.143)^2 \times \frac{0.001}{0.30} + (4 - 0.143)^2 \times \frac{0.00}{0.30} \approx 0.209 \end{aligned}$$

- Windows computer:

$$\text{var}(C \mid OS=0) = \sum_{i=1}^4 \left[(c_i - E(C \mid OS=0))^2 P(C = c_i \mid OS=1) \right] \approx 0.809$$

Several Discrete Random Variables

Covariance and Correlation

► **Covariance** between X and Y :

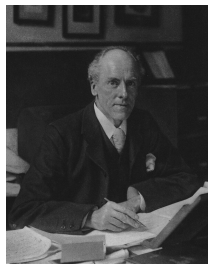
$$\text{cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y$$

- $\sigma_{XY} > 0$: positive association (when $X > \mu_X$ then $Y > \mu_Y$, and when $X < \mu_X$ then $Y < \mu_Y$)
- $\sigma_{XY} < 0$: negative association
- $\sigma_{XY} = 0$: neither negative nor positive association.

► **Correlation** between X and Y :

$$\text{corr}(X, Y) = \rho = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$$

- Pearson product-moment correlation coefficient
- unitless measure
- $-1 \leq \rho \leq 1$



Karl Pearson
1857–1936

Continuous Random Variables

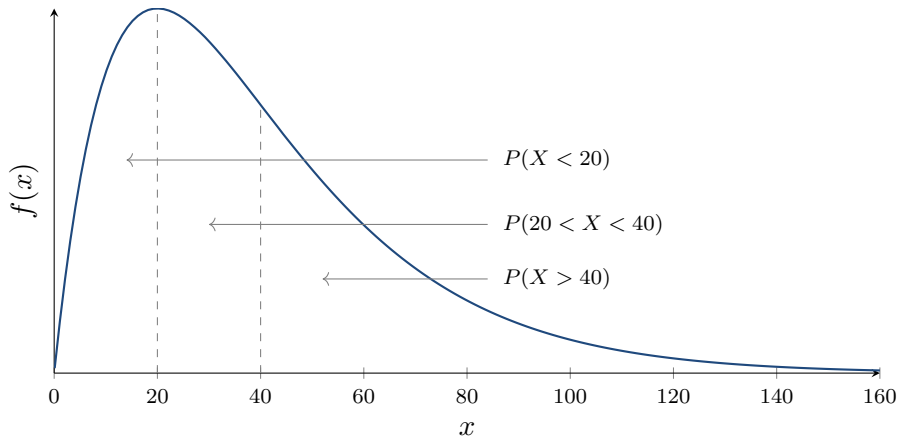
- ▶ Continuous random variables can take any value in an interval
 - GDP, interest rates, income, stock-market indices, exchange rates are all (treated as) continuous variables
- ▶ Because they can take uncountably many values, the probability that any single value occurs is zero
 - e.g. if X is tomorrow's EUR|CHF exchange rate, $P(X = 0.9300) = 0$
- ▶ For continuous variables, probability statements are meaningful only when we consider outcomes within intervals
 - e.g. $P(0.9295 \leq X \leq 0.9305)$ is meaningful ("0.9300" is really an interval)

Continuous Random Variables

Probability Density Function

- ▶ The **probability density function (pdf)** summarizes the probabilities of a continuous variable
- ▶ Let X be a continuous random variable with pdf $f(x)$:
 - $f(x) \geq 0$
 - $\int_{-\infty}^{\infty} f(x) dx = 1$
 - $P(a \leq X \leq b) = \int_a^b f(x) dx$
- ▶ The probability that X falls in the interval $[a, b]$ equals the area under the pdf between these points.

pdf of a Continuous Random Variable



Hill, Griffiths, & Lim (2008)

Continuous Random Variables

Cumulative Density Function

- ▶ The **cumulative distribution function (cdf)** is defined as in the discrete case: the probability that the random variable is less than or equal to a value:

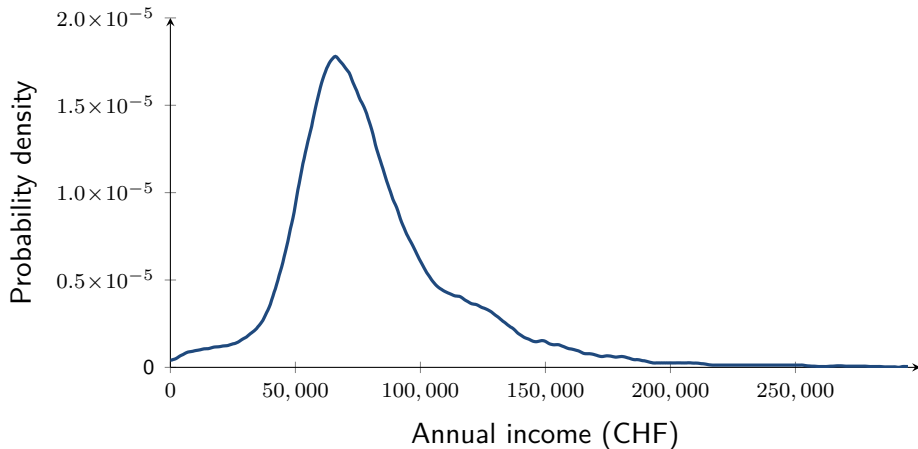
$$P(X \leq a) = \int_{-\infty}^a f(x) dx = F(a)$$

- ▶ the cdf is obtained by *integrating* the pdf
- ▶ hence the pdf can be obtained by *differentiating* the cdf:

$$f(x) = \frac{dF(x)}{dx} = F'(x)$$

An Empirical *pdf*

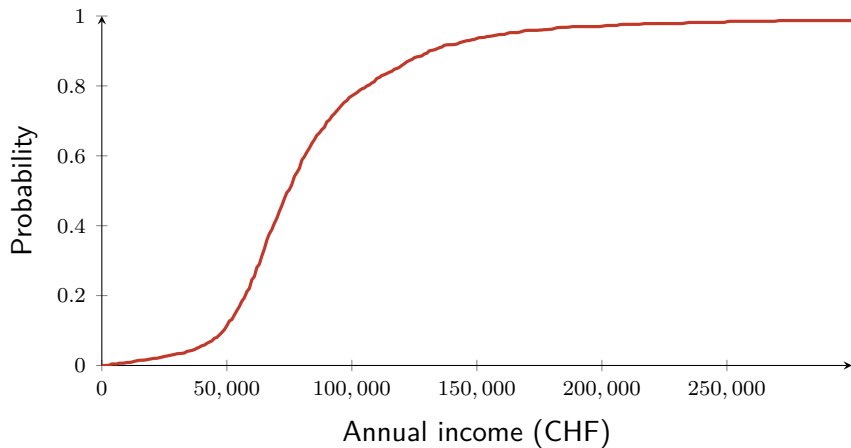
Male Workers' Income in Switzerland



Data: Swiss Labour Force Survey 2008

An Empirical *cdf*

Male Workers' Income in Switzerland



Data: Swiss Labour Force Survey 2008

Continuous Random Variables

Expectation and Variance

► Expectation:

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

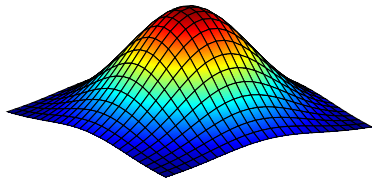
- Compared to the discrete case, the integral simply replaces the sum
- Interpretation: the average value of X over an infinite number of draws

► Variance:

$$\sigma_X^2 = E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx = E(X^2) - \mu_X^2$$

Several Continuous Random Variables

Joint, Marginal, and Conditional Probability Distributions



- ▶ The joint probability density function $f(x, y)$ is a *surface*; probabilities are *volumes* under the surface

- ▶ Probability that $X \in [a, b]$ **and at the same time** $Y \in [c, d]$:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_{x=a}^b \int_{y=c}^d f(x, y) dx dy$$

- ▶ Marginal probability density function for Y :

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

- ▶ Conditional probability density function:

$$f(y | x) = \frac{f(x, y)}{f(x)}$$

→ unlike the discrete case, $f(y | x)$ is not a probability but a density used to compute probabilities:

$$E(Y | X = x) = \int_{-\infty}^{\infty} y f(y | x) dy$$

Normal Distribution

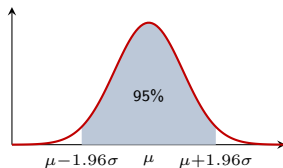
- ▶ **Normal (Gaussian) distribution:** continuous probability distribution with a bell-shaped pdf
- ▶ A normally distributed random variable X with mean μ and variance σ^2 is written

$$X \sim N(\mu, \sigma^2)$$

- ▶ pdf of X :

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- ▶ **Standard normal distribution:** $Z \sim N(0, 1)$
 - tables/software give the standard normal cdf, usually denoted $\Phi(z) = P(Z \leq z)$
 - **standardization** to compute probabilities for $X \sim N(\mu, \sigma^2)$: $z = \frac{x - \mu}{\sigma}$



Abraham de Moivre
1667–1754

Normal Distribution (continued)

Calculating probabilities

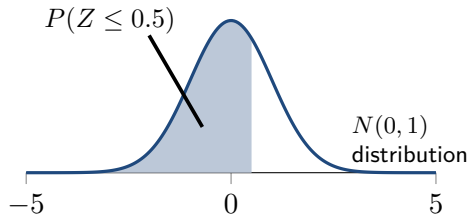
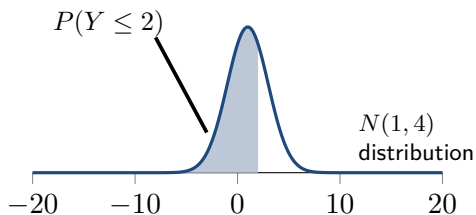
- What is the probability that $Y \leq 2$ if $Y \sim N(1, 4)$?

- Step 1: z -standardize to get the z -score

$$z = \frac{x - \mu}{\sigma} = \frac{2 - 1}{2} = 0.5$$

- Step 2: compare the z -score to the distribution table

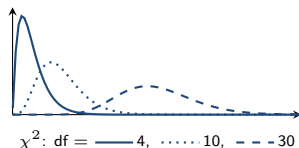
$$P\left(\frac{Y - 1}{2} \leq 0.5\right) = \Phi(0.5) = 0.691$$



Other important distributions

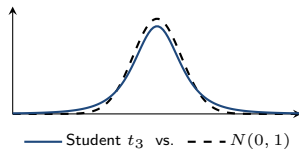
► Chi-square distribution

- $V = Z_1^2 + Z_2^2 + \dots + Z_m^2 \sim \chi_{(m)}^2$
with m degrees of freedom
- where $Z_i \sim N(0, 1)$, $i = 1, \dots, m$, and the Z_i are independent
- as $m \rightarrow \infty$, $\chi_{(m)}^2 \rightarrow Z \sim N(0, 1)$, and $\chi_{(1)}^2 = Z_1^2$..



► Student t -distribution

- $t = \frac{Z}{\sqrt{V/m}} \sim t_{(m)}$ with m degrees of freedom
- where $Z \sim N(0, 1)$ and $V \sim \chi_{(m)}^2$
- as $m \rightarrow \infty$, $t_{(m)} \rightarrow Z \sim N(0, 1)$



► F -distribution

- $F = \frac{V_m/m}{V_n/n} = \frac{\chi_m^2}{m} \cdot \frac{n}{\chi_n^2} \sim F_{(m,n)}$
with m, n degrees of freedom
- relationship to χ^2 : $\lim_{n \rightarrow \infty} F = \frac{\chi_m^2}{m}$

